

RISHII KUMAR SINGH

AI/ML Engineer and Undergraduate Researcher

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PROFESSIONAL SUMMARY

AI/ML engineer and undergraduate researcher building evaluated systems across computer vision, multilingual NLP, AI agents, and full-stack applications. Experience with defense imagery, open-source agent infrastructure, retrieval-augmented generation, reproducible evaluation, and deployment using Python, PyTorch, FastAPI, React, Docker, and ONNX.

EDUCATION

Bachelor of Technology in Environmental Engineering | *Delhi Technological University* **2025 to 2029**

SGPA: 8.90/10 after Semester 1 | Independent focus: AI/ML, NLP, knowledge graphs, AI agents, computer vision, and full-stack engineering

CBSE Class XII, Science with Computer Science | *Navy Children School* **2025**

Score: 85.2%

EXPERIENCE

Research Intern | *AIMS-DTU*

Jun 2026 to Present

- Selected through multiple interview rounds for a supervised research internship under Prof. Dinesh K. Vishwakarma, working toward a research manuscript and conference submission.
- Conduct literature review, data validation, experiment design, baseline evaluation, systematic model analysis, and reproducible documentation using version-controlled research pipelines.

Open Source Contributor | *GirlScript Summer of Code*

May 2026 to Aug 2026

- Ranked in the top 4% globally, placing 1,605 of 43,587 participants with 2,507 points and 13 merged pull requests across Heli-OS, AegisAI, and Sahidawa India.
- Completed six advanced and six intermediate contributions spanning distributed agent execution, context compression, crash recovery, WebSocket streaming, race-condition prevention, and dependency-aware parallelization.

AI/ML Intern | *WESEE, Indian Navy*

Jun 2026 to Jul 2026

- Detected 123 duplicate satellite images across training and evaluation splits, rebuilt a leakage-free YOLO11x-ORB pipeline, and improved vessel detection mAP50 from 0.434 to 0.921 and mAP50-95 from 0.111 to 0.661.
- Completed eight human-reviewed active-learning rounds, exported models through ONNX, trained UNet++ coastal segmentation to 0.853 mIoU, and achieved 97.1% ship-type classification accuracy.

AI/ML Intern, Technical Development | *Artha Research and Intelligence Lab*

Oct 2025 to Dec 2025

- Built a Python and QGIS road-network accessibility pipeline covering more than 500 healthcare facilities across 12 districts of Himachal Pradesh.
- Identified and mapped more than 30 underserved regions using road, population, terrain, and hazard information to support healthcare-access prioritization.

SELECTED PROJECTS

PolyAgent CI | **Multi-Agent Orchestration** | [GitHub](#)

Apr 2026

- Built a Python orchestrator that coordinates four coding agents across isolated Git worktrees and schedules tasks through a dependency-aware directed acyclic graph.
- Implemented contract-first development, semantic review, automated conflict resolution, test-gated topological merges, 14 DAG tests, and collaborative editor synchronization under 200 ms.

Nyaya Navigator | **Offline Multilingual Legal Navigation** | [GitHub](#)

Jul 2026

- Built an offline English, Hindi, and Hinglish legal-navigation system over 6,845 official-law chunks using lexical retrieval and local embeddings.
- Implemented fact confirmation, claim-level source verification, effective-date handling, refusal, escalation, OCR, speech, and a loopback-only FastAPI and React application.

Digital Twin of Nikola Tesla | **Grounded RAG** | [GitHub](#)

Jun 2026

- Built and deployed a conversational digital twin grounded in Tesla's writings, patents, and interviews with passage retrieval, source citations, and locally generated embeddings stored in ChromaDB.
- Added short-term and persistent memory, streamed responses, confidence labels, PDF ingestion, voice interaction, an interactive knowledge graph, quizzes, and a Tesla-versus-Edison debate mode.

Jarvis | **Voice and Gesture Desktop Control** | [GitHub](#)

Feb 2026

- Built a multimodal desktop-control system combining voice commands, real-time hand tracking, ten built-in gestures, and user-defined gesture training.
- Trained a KNN classifier on 3,261 samples with 99.54% measured accuracy and under 50 ms latency, with a Flask and Socket.IO dashboard for data collection, training, and process control.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL

AI and ML: PyTorch, TensorFlow, Scikit-Learn, Hugging Face, OpenCV, MediaPipe, ONNX, RAG, NLP, LLMs, object detection, image segmentation, active learning

Engineering: FastAPI, Flask, React, PostgreSQL, SQLite, ChromaDB, REST APIs, WebSocket, Docker, Git, GitHub, Linux, QGIS

ACHIEVEMENTS AND CERTIFICATIONS

Achievements: Winner, Green Tag Case Competition at BITS Pilani APOGEE 2026; IOQM Certificate of Merit, top 10%; qualified the UPSC NDA and NA written examination three times.

Certifications: Deep Learning Specialization and Machine Learning Specialization, DeepLearning.AI; Data Science and Artificial Intelligence, CODE IIT Madras.